2D U-Net Segmentation of Multitarget Deep Brain Stimulation Lead Trajectories on Postoperative Imaging

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Background

- Deep brain stimulation (DBS) can be an effective therapy for managing symptoms associated with movement disorders and neuropsychiatric conditions
- Despite DBS benefiting many patients, there is still significant unexplained outcome variance
- Lead placement has shown to influence outcomes, **[1,2]** however, there may be potential side effects arising from full implant trajectories **[3,4]**
- There is a historic lack of methods to map lead trajectories causing lead trajectory effects to remain largely unexplored

Objectives

- **Develop an automated, open-source tool for DBS lead segmentation from postop CT/MRI**
- Later, leverage the tool to investigate lead trajectory effects on DBS clinical outcomes

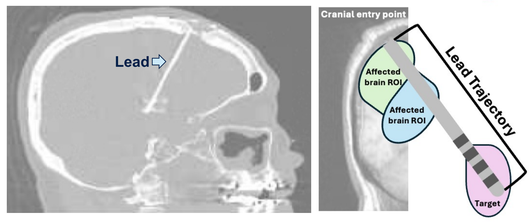


Figure 1: *Left.* Postop CT with lead artifact illustrating segmentation problem. *Right.* Lead trajectory schematic.

Cohort & Model

- Postop CTs collected from **138 subjects** with DBS for a movement disorder
- Ground truth labeling of lead in ITK-Snap → 3D binary label masks
- 3D CT volumes → skull stripped using HD-CTBET → z-score 2D slices (512x512 pixels)
- CNN implemented using PyTorch
- 2D U-Net model for slice-wise segmentation
- Combination of weighted dice loss (30%), focal loss (30%), binary cross entropy (40%)

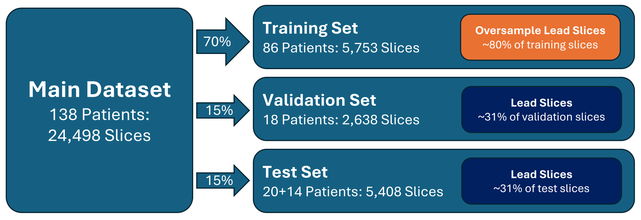


Figure 2A: Data split strategy. Lead slices were oversampled in training set to address the class imbalance. [Test Set recently increased by 14]

Cohort and Model Cont.

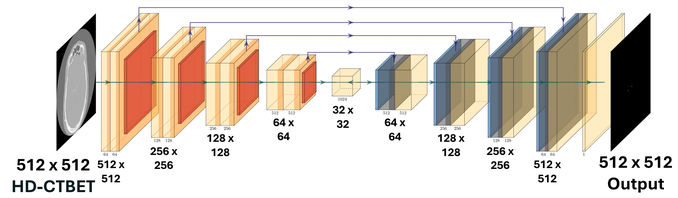


Figure 2B: U-Net Architecture from CT Input to Segmentation Output.

Results

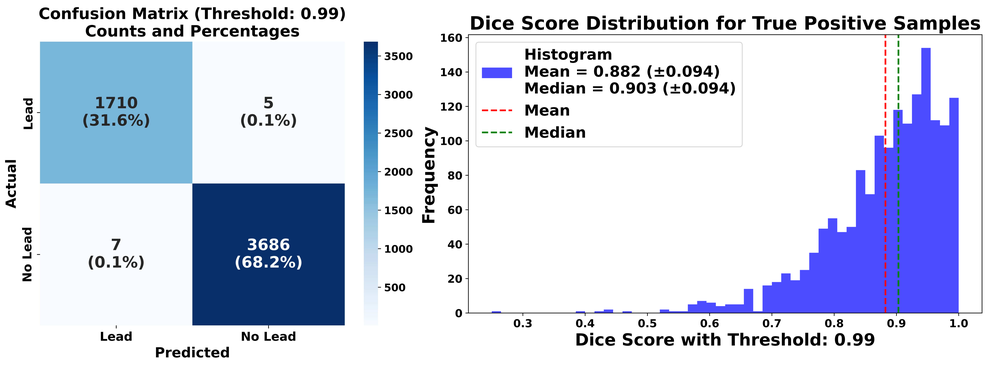


Figure 3: U-Net model performance with threshold of 0.99. *Left.* Confusion matrix demonstrating counts and percentages of True Positive, True Negative, False Positive, and False Negative predictions on 5,408 test slices. *Right.* True Positive Dice Score distribution.

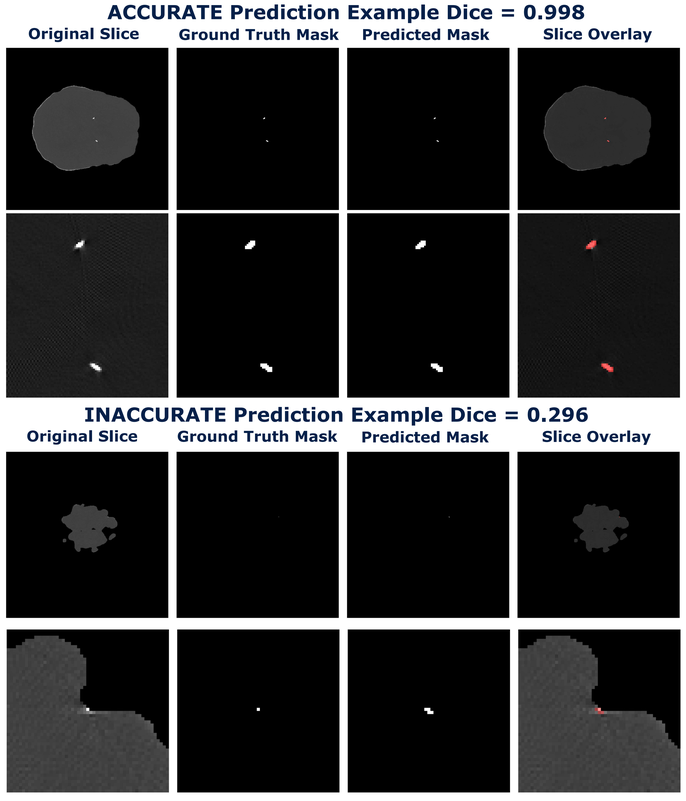


Figure 4: Representative 2D examples of model performance. *Top.* Accurate segmentation example. *Bottom.* Inaccurate segmentation example, spatially located near the edge of the brain.

Results Cont.

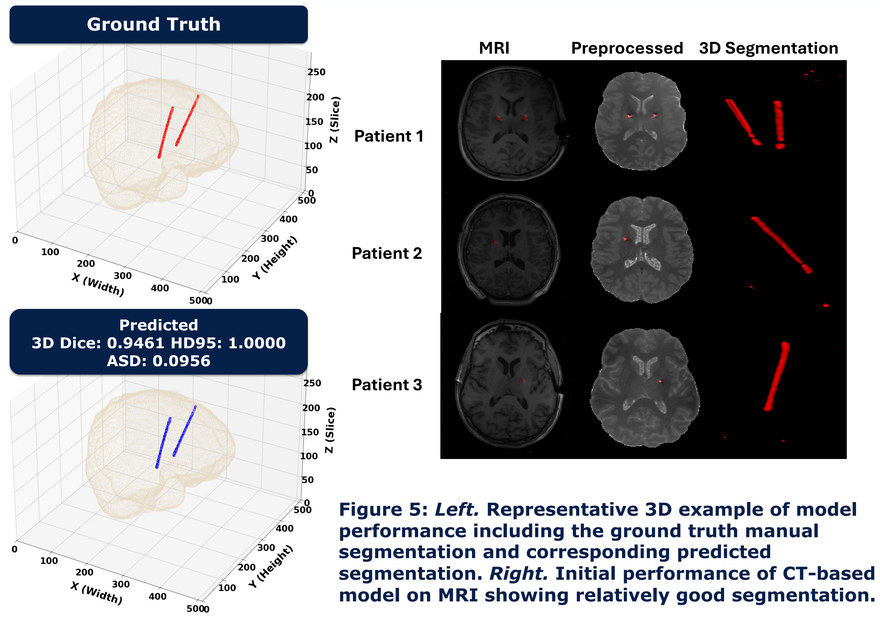


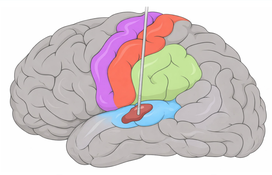
Figure 5: *Left.* Representative 3D example of model performance including the ground truth manual segmentation and corresponding predicted segmentation. *Right.* Initial performance of CT-based model on MRI showing relatively good segmentation.

Discussion & Limitations

- Our model achieved robust segmentation with **slice- and volume-based True Positive Mean Dice scores of 0.8820±0.094 and 0.8865 ±0.0564**, Mean HD95 (voxels): 1.0122, Mean ASD (voxels): 0.1901
- DBS lead pixels accounting for on average 0.01% of each slice, yield a high class imbalance

Next Steps

- To further enhance performance, we will expand the dataset, apply data augmentation, and advance class imbalance solutions
- CT-to-MRI transfer learning will enable robust MRI-based segmentation
- Atlas mapping of brain region to lead trajectory



[1 0 1 0 1 1 0 1]
Brain Area Binary Vector

Acknowledgments

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References

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